FRONT-END DEVELOPMENT

HACKATHON

Level-2

STUDENTNAME : ANANDHI K
YEAR : II
COLLEGE CODE : BRUAU
COLLEGE NAME : GOVERNMENT ARTS AND SCIENCE COLLEGE (CO-ED) AVINASHI
DEPARTMENT : B.Sc (COMPUTER SCIENCE)
SEMESTER : IV
ROLL NO : 2422K1154
NMID : 7C2104E5C973CFB8FF649439409F7B6E

DATE : 13/04/2026
TITLE : STUDENT ACTIVITY TRACKER WITH REAL-TIME PROGRESS AND CERTIFICATE GENERATION
GITHUB LINK :https://github.com/anandhikannan2007-boop/Hackathon-level-2.git

1. OBJECTIVE

The main objective of this project is to implement front-end interactivity to track student activities, update their completion status, and display real-time progress using JavaScript.

2. PROBLEM STATEMENT

In Level-1, the UI was static and did not respond to user actions. In Level-2, the focus is on converting it into an interactive UI where user actions like clicking buttons update activity status and progress dynamically using event handling.

3. KEY FEATURES IMPLEMENTED

Mark activities as completed
Real-time progress tracking
Sound effect on button click
Certificate download after completion

4. UI COMPONENT DESIGN

The application includes key UI components such as Login Page, Activity List, Done Buttons, Progress Bar, Navigation Menu, and Certificate Download Button. The design is simple and user-friendly.

5. TECNOLOGY USED

HTML – Structure of the application
CSS – Styling and animations
JavaScript – Logic and interactivity

6. DATA HANDLING APPROACH

The activity data is stored in a JavaScript array. LocalStorage is used to save user data and activity status so that the data persists even after refreshing the page.

7.APPLICATION OVERVIEW

This is a web-based Activity Tracker application. Users can log in, mark activities as completed, view their progress, and download a certificate after completing all tasks.

8. INTERACTIVITY IMPLEMENTATION

When the user clicks the “Done” button, an event is triggered that updates the activity status to completed. The UI is dynamically updated without reloading the page.

9.APPLICATION LOGIC

The logic counts the number of completed activities and compares it with the total number of activities. Based on this, the progress percentage is calculated and displayed using a progress bar.

10.BEST PRACTICE FOLLOWED

Meaningful variable names are used. Code is kept clean and readable. LocalStorage is used efficiently. Reusability and proper structure are maintained.

11.EXPECTED OUTPUT / DELIVERABLES

- A fully working interactive web application
- Activity tracking system
- Real-time progress display
- Certificate generation feature
- Well-structured GitHub source code

12.LEARNING OUTCOME

Learned JavaScript event handling
Improved DOM manipulation skills
Understood LocalStorage usage
Gained experience in building interactive UI

13.CONCLUSION

This application successfully implements front-end interactivity and improves user experience by providing real-time updates and dynamic UI behavior.

GITHUB LINK

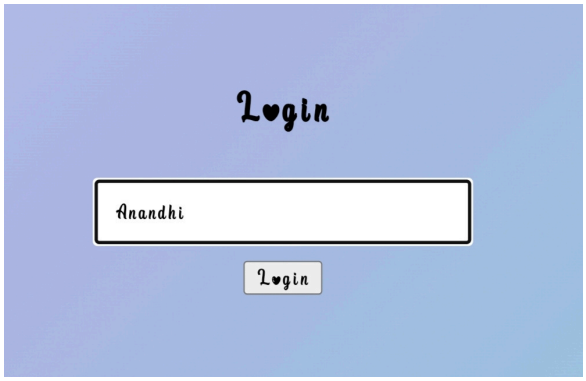
https://github.com/anandhikannan2007-boop/Hackathon-level-2.git

OUTPUT

STEP 1

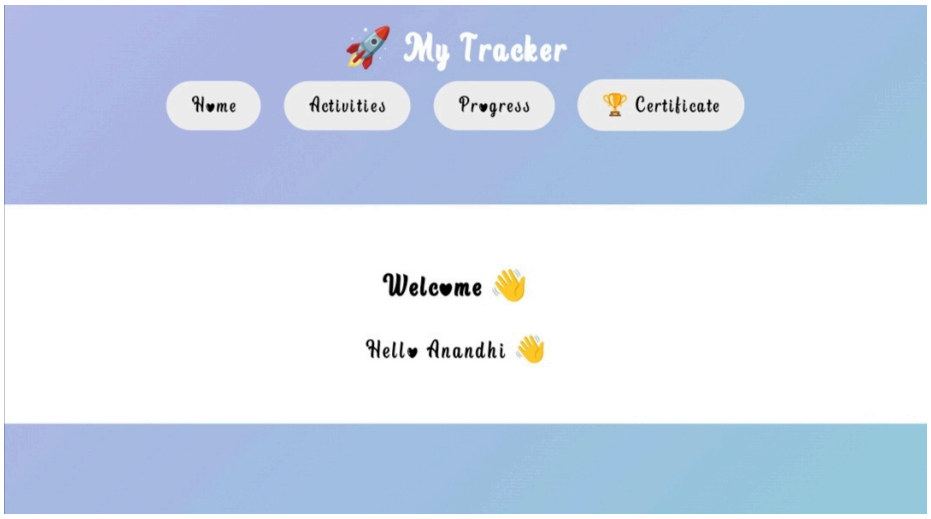
Page loads and login screen is displayed

User enters name and clicks login



STEP 2

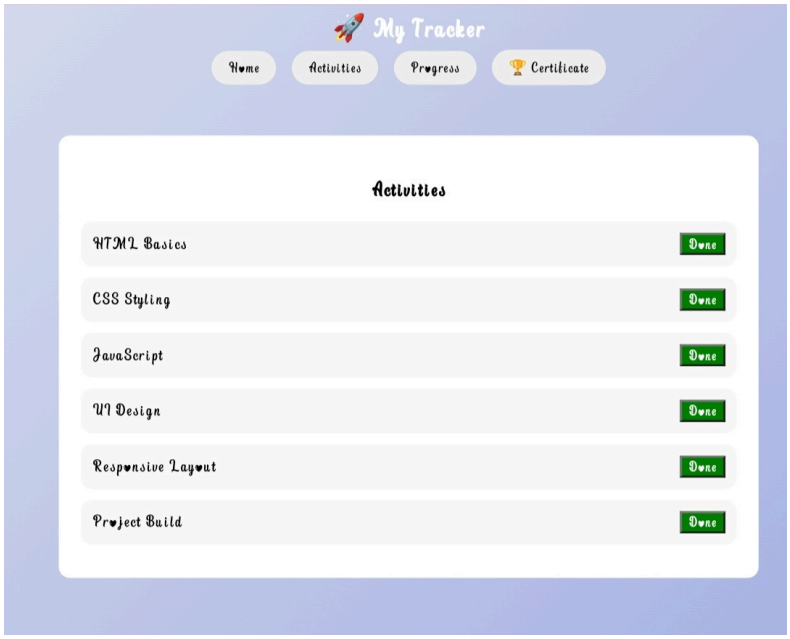
Welcome message with user name is displayed



.

STEP 3

Activity list is loaded from localStorage
users views activities and click " Done "

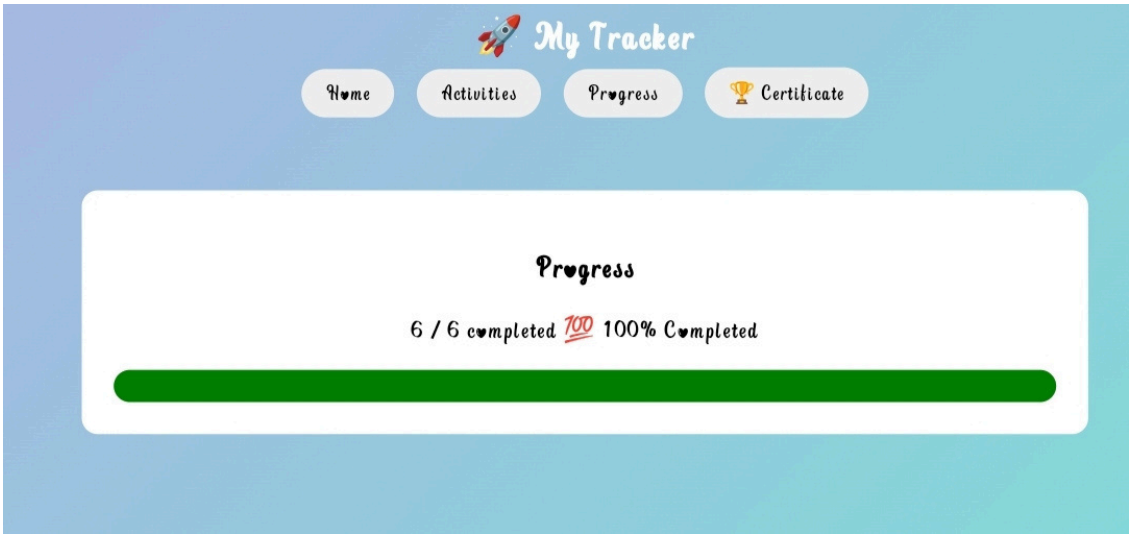


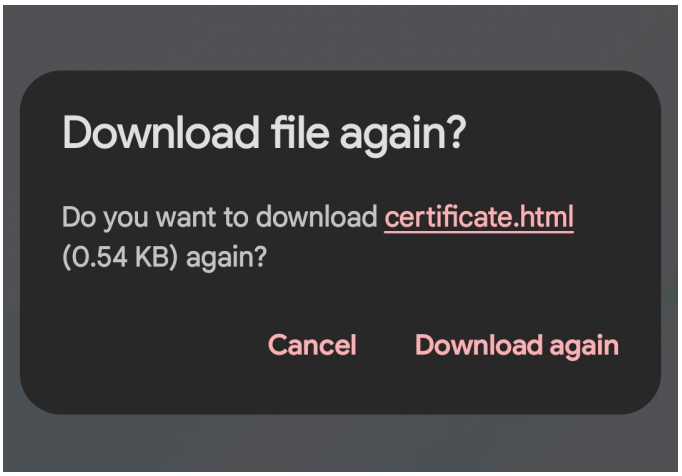
STEP 4

Activity is marked as completed and sound plays

STEP 5

Progress page shows completed count and progress bar
If all tasks completed, 100 % message shown



STEP 6

User clicks certificate button and file is downloaded